

Section-A

Units and Measurements

(Dimensions of Physical Quantities)

1. The quantities which have the same dimensions as those of solid angle are:
- strain and arc
 - angular speed and stress
 - strain and angle
 - stress and angle

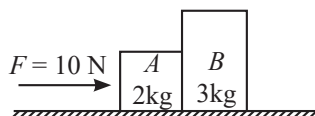
(Measuring Instruments)

2. In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:
- 100 N
 - $10(N + 1)$
 - $\frac{1}{10N}$
 - $\frac{1}{100(N + 1)}$

Laws of Motion

(Motion of Connected Bodies)

3. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is:



- 6 N
- 10 N
- Zero
- 4 N

(Circular Motion, Banking of Road)

4. A particle moving with uniform speed in a circular path maintains

- constant velocity by varying acceleration.
- varying velocity and varying acceleration.
- constant velocity.
- constant acceleration.

5. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T . If speed becomes 20ω while keeping the same radius, the tension in the string becomes:

- $\frac{T}{4}$
- $\sqrt{2}T$
- T
- $4T$

Work, Energy and Power

(Power)

6. At any instant of time t , the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of 5 N. The value of instantaneous power is (in SI unit):

- 7
- 6
- 10
- 5

(Collision)

7. Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2 . The ratio $v_1 : v_2$ is:
- 4 : 1
 - 1 : 4
 - 1 : 2
 - 2 : 1

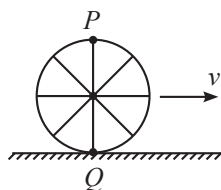
System of Particles and Rotational Motion

(Moment of Inertia, Theorem of Parallel and Perpendicular Axis and Energy in Rotation)

8. The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is 2400 g cm². The length of the 400 g rod is nearly:
- 20.7 cm
 - 72.0 cm
 - 8.5 cm
 - 17.5 cm

(Rolling Motion)

9. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



- Both the points P and Q move with equal speed.
- Point P has zero speed.
- Point P moves slower than point Q .
- Point P moves faster than point Q .

Gravitation

(Newton's Law of Gravitation & Acceleration Due to Gravity)

10. The mass of a planet is $\frac{1}{10}$ th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is:
- 4.9 m s^{-2}
 - 3.92 m s^{-2}
 - 19.6 m s^{-2}
 - 9.8 m s^{-2}

Mechanical Properties of Solids

(Stress, Strain and Young's Modulus)

11. The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8 \text{ N m}^{-2}$ and $2 \times 10^{11} \text{ N m}^{-2}$, is:
- 40 mm
 - 8 mm
 - 4 mm
 - 0.4 mm

Mechanical Properties of Fluid

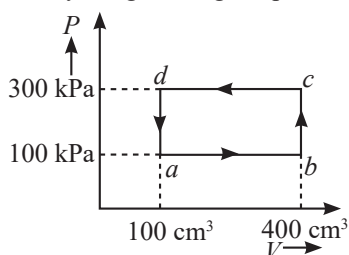
(Surface Tension and Surface Energy)

12. A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 Nm^{-1} , then the excess force required to take it away from the surface is :
- 1.98 mN
 - 99 N
 - 19.8 mN
 - 198 N

Thermodynamics

(Thermodynamic Processes and First Law of Thermodynamics)

13. A thermodynamic system is taken through the cycle $abcd$. The work done by the gas along the path bc is:



- 90J
- 60J
- Zero
- 30J

Oscillations

(Time Period and Frequency)

14. If $x = 5 \sin\left(\pi t + \frac{\pi}{3}\right)$ m represents the motion on a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are:
- 5 cm, 1 s
 - 5 m, 1 s
 - 5 cm, 2 s
 - 5 m, 2 s

Electrostatic Potential and Capacitance

(Electric Potential)

15. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, $4 \times 10^{-6} \text{ C m}$, is $\pm 9 \times 10^3 \text{ V}$.

Reason R: $\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI Units}\right)$

Reason R: $V = \pm \frac{2P}{4\pi\epsilon_0 r^2}$, where r is the distance of any

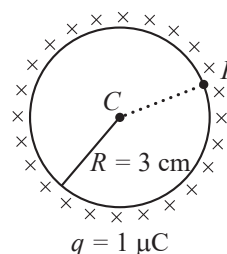
axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the correct answer from the options given below:

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true and R is NOT the correct explanation of A.

(Potential Difference and Work Done)

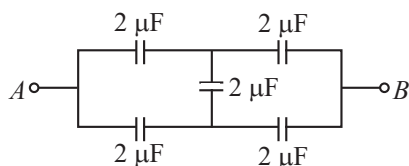
16. A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the given is: $\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}\right)$



- 0.5×10^5
- zero
- 3×10^5
- 4×10^5

(Grouping of Capacitors)

17. In the following circuit, the equivalent capacitance between terminal A and terminal B is:



- a. $0.5 \mu\text{F}$ b. $4 \mu\text{F}$
c. $2 \mu\text{F}$ d. $1 \mu\text{F}$

Current Electricity

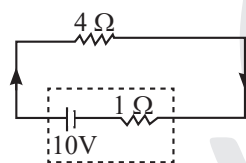
(Combination of Resistance)

18. A wire of length ' l ' and resistance 100Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

- a. 55Ω b. 60Ω
c. 26Ω d. 52Ω

(Internal Resistance of a Cell and Grouping)

19. The terminal voltage of the battery, whose emf is $10V$ and internal resistance 1Ω , when connected through an external resistance of 4Ω as shown in the figure is :



- a. $8 V$ b. $10 V$
c. $4 V$ d. $6 V$

Moving Charges and Magnetism

(Magnetic Field in Circular Coil)

20. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

- a. 4.4 mT b. 44 T
c. 44 mT d. 4.4 T

Magnetism and Matter

(Magnetic Properties of Materials)

21. Match List-I with List-II.

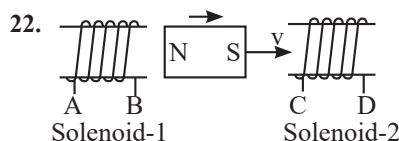
	List-I (Material)		List-II (Susceptibility (χ))
A.	Diamagnetic	I.	$\chi = 0$
B.	Ferromagnetic	II.	$0 > \chi \geq -1$
C.	Paramagnetic	III.	$\chi \gg 1$
D.	Non-magnetic	IV.	$0 < \chi < \epsilon$ (a small positive number)

Choose the correct answer from the options given below:

- a. A-III, B-II, C-I, D-IV
b. A-IV, B-III, C-II, D-I
c. A-II, B-III C-IV, D-I
d. A-II, B-I, C-III, D-IV

Electromagnetic Induction

(Lenz's Law)

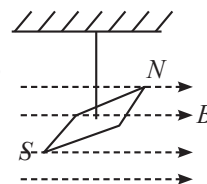


In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

- a. AB and CD b. BA and DC
c. AB and DC d. BA and CD

(Dipole)

23. In a uniform magnetic field of 0.049 T , a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is $9.8 \times 10^{-6} \text{ kg m}^2$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5} \text{ Am}^2$; then the value of ' x ' is:



- a. $50 \pi^2$ b. $1280 \pi^2$
c. $5 \pi^2$ d. $128 \pi^2$

Alternative Current

(Transformer)

24. In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$. The ratio

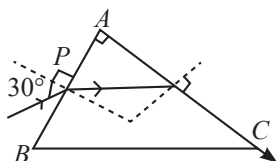
$V_s : V_p$ is equal to (the symbols carry their usual meaning):

- a. 1:1 b. 1:4
c. 1:2 d. 2:1

Ray Optics and Optical Instruments

(Refraction Through Prism and Dispersion Through Prism)

25. A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC . The refractive index of the prism is:



- a. $\frac{\sqrt{3}}{4}$
 b. $\frac{\sqrt{3}}{2}$
 c. $\frac{\sqrt{5}}{4}$
 d. $\frac{\sqrt{5}}{2}$

Wave Optics

(Interference and Young's Double Slit Experiment)

26. If the monochromatic source in Young's double slit experiment is replaced by white light, then
- there will be a central bright white fringe surrounded by a few coloured fringes.
 - all bright fringes will be of equal width.
 - interference pattern will disappear.
 - there will be a central dark fringe surrounded by a few coloured fringes.

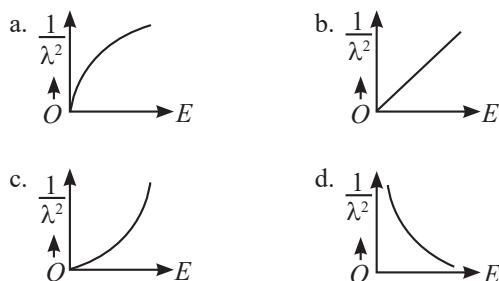
(Polarization Phenomenon, Law of Malus and Brewster's Law)

27. An unpolarised light beam strikes a glass surface at Brewster's angle. Then
- both the reflected and refracted light will be completely polarised.
 - the reflected light will be completely polarised but the refracted light will be partially polarised.
 - the reflected light will be partially polarised.
 - the refracted light will be completely polarised.

Dual Nature of Radiation and Matter

(Wave Nature of Matter [De-Broglie Wavelength])

28. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):



Parameters of Photon (Momentum, Pressure and Energy)

29. If c is the velocity of light in free space, the correct statements about photon among the following are :

- The energy of a photon is $E = h\nu$
 - The velocity of a photon is c .
 - The momentum of a photon, $p = \frac{h\nu}{c}$
 - In a photon-electron collision, both total energy and total momentum are conserved.
 - Photon possesses positive charge.
- Choose the correct answer from the options given below:
- A, C and D only
 - A, B, D and E only
 - A and B only
 - A, B, C and D only

Atoms

(Alpha-Particle Scattering and Rutherford's Nuclear Model of Atom [Distance of Closest Approach and Impact Parameter, Electron Orbits])

30. Given below are two statements:

Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II: Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

(Bohr H-atom Model, Radius of Orbit, Velocity and Energy of Electrons, Wavelengths of Hydrogen Spectrum and Ionisation)

31. Match List I with List II.

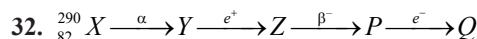
List I (Spectral Lines of Hydrogen for transitions from)		List II (Wavelengths (nm))	
A.	$n_2 = 3$ to $n_1 = 2$	I.	410.2
B.	$n_2 = 4$ to $n_1 = 2$	II.	434.1
C.	$n_2 = 5$ to $n_1 = 2$	III.	656.3
D.	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-I, B-II, C-III, D-IV
- A-II, B-I, C-IV, D-III
- A-III, B-IV, C-II, D-I

Nuclei

(Alpha, Beta and Gamma Decay)



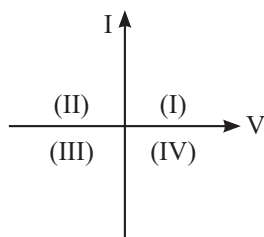
In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are:

- a. 288, 82 b. 286, 81
c. 280, 81 d. 286, 80

Semiconductor Electronics

(Optoelectronic Devices)

33. Consider the following statement A and B and identify the correct answer:



- A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased *pn* junction diode, the current measured in (A), is due to majority charge carriers.

- a. Both A and B are correct.
b. Both A and B are incorrect.
c. A is correct but B is incorrect.
d. A is incorrect but B is correct.

(Logic gate)

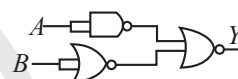
34. A logic circuit provides the output Y as per the following truth table:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is:

- a. $\bar{B}$ b. B
c. $A.B + \bar{A}$ d. $A.\bar{B} + \bar{A}$

35. The output (Y) of the given logic gate is similar to the output of an/a:



- a. OR gate b. AND gate
c. NAND gate d. NOR gate

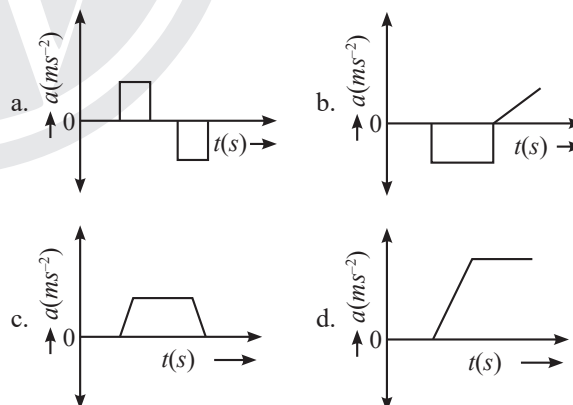
SECTION-B

Units and Measurements

(Dimensions of Physical Quantities)

36. A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at given time t . The factor which is dimensionless, if α and β are constants, is:

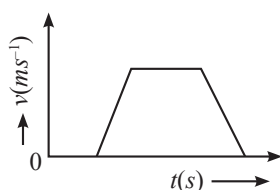
- a. $\alpha\beta t$
b. $\alpha\beta/t$
c. $\beta t/\alpha$
d. $\alpha t/\beta$



Motion in a Straight Line

(Graphs)

37. The velocity (v) – time (t) plot of the motion of a body is shown below:



The acceleration (a) – time (t) graph that best suits this motion is:

Gravitation

(Satellite, Orbital Velocity and Escape Velocity)

38. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

- a. $\frac{GmM}{2R}$ b. $\frac{GmM}{3R}$
c. $\frac{5GmM}{6R}$ d. $\frac{2GmM}{3R}$

Thermal Properties of Matter

(Thermal Expansion)

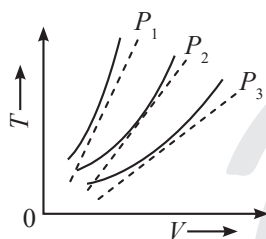
39. A metallic bar of Young's modulus, $0.5 \times 10^{11} \text{ Nm}^{-2}$ and coefficient of linear thermal $10^{-5} \text{ } ^\circ\text{C}^{-1}$ length 1m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

- a. $100 \times 10^3 \text{ N}$ b. $2 \times 10^3 \text{ N}$
c. $5 \times 10^3 \text{ N}$ d. $50 \times 10^3 \text{ N}$

Kinetic Theory

(Ideal Gas Equations and Vander Waals Relation)

40. The following graph represents the T - V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1 , P_2 and P_3 compared with those of Charles's law represented as dotted lines.



- a. $P_2 > P_1 > P_3$ b. $P_1 > P_2 > P_3$
c. $P_3 > P_2 > P_1$ d. $P_1 > P_3 > P_2$

Oscillations

(Simple Pendulum and Loaded Springs)

41. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original

length, then the new time period of oscillation is $\frac{x}{2}$ time its original time period. Then the value of x is:

- a. $2\sqrt{3}$ b. 4
c. $\sqrt{3}$ d. $\sqrt{2}$

42. If the plate of a parallel plate capacitor connected to a battery are moved close to each other, then

- A. the charge stored in it. increases.
B. the energy stored in it. decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.

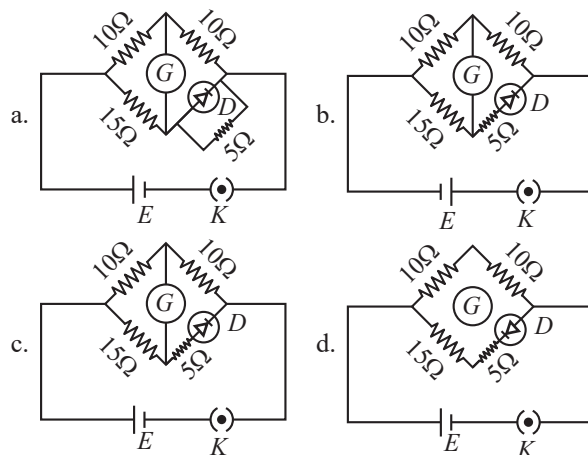
Choose the most appropriate answer from the given below:

- a. B, D and E only b. A, B and C only
c. A, B and E only d. A, C and E only

Current Electricity

(Wheatstone and Meter Bridge)

43. Choose the correct circuit which can achieve the bridge balance.



(Electrical Power)

44. Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

- a. 1 : 2 b. 2 : 3
c. 1 : 1 d. 2 : 9

Moving Charges and Magnetism

(Magnetic Moment)

45. An iron bar of length L has magnetic moment M . It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is:

- a. $2M$ b. $\frac{M}{\sqrt{3}}$
c. M d. $\frac{M}{2}$

Magnetism and Matter

(Magnetic Dipole Moment, and Force Between Magnetic Poles 60°)

46. A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

- A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

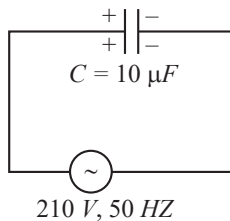
Choose the correct statement(s) from the options given below:

- a. A, C and D only b. C only
c. B and D only d. A and C only

Alternating Current

(A.C. Circuit Containing Capacitance Only)

47. A 10mF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):



- a. 1.20 A
- b. 0.35 A
- c. 0.58 A
- d. 0.93 A

- a. they travel with speed equal to $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$.
- b. they originate from charges moving with uniform speed.
- c. they are transverse in nature.
- d. the energy density in electric field is equal to energy density in magnetic field.

(Displacement Current)

49. A parallel plate capacitor is charged by connecting it to a battery/through a resistor. If I is the current in the circuit, then in the gap between the plates:
- a. displacement current of magnitude equal to I flows in a direction opposite to that of I .
 - b. displacement current of magnitude greater than I flows but can be in any direction.
 - c. there is no current.
 - d. displacement current of magnitude equal to I flows in the same direction as I .

Electromagnetic Waves

(Properties and Applications [i.e. Velocity, Amplitude, Energy Density] of Electromagnetic Waves)

48. The property which is not of an electromagnetic wave travelling in free space is that:

Ray Optics and Optical Instruments

(Telescope)

50. A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:
- a. 17
 - b. 32
 - c. 34
 - d. 28

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (a) | 4. (b) | 5. (d) | 6. (c) | 7. (d) | 8. (c) | 9. (d) | 10. (b) |
| 11. (c) | 12. (c) | 13. (c) | 14. (d) | 15. (a) | 16. (b) | 17. (c) | 18. (d) | 19. (a) | 20. (a) |
| 21. (c) | 22. (c) | 23. (b) | 24. (d) | 25. (d) | 26. (a) | 27. (b) | 28. (b) | 29. (d) | 30. (a) |
| 31. (d) | 32. (b) | 33. (c) | 34. (a) | 35. (b) | 36. (d) | 37. (a) | 38. (c) | 39. (d) | 40. (b) |
| 41. (d) | 42. (d) | 43. (c) | 44. (d) | 45. (d) | 46. (d) | 47. (d) | 48. (b) | 49. (d) | 50. (d) |

Explanations

1. (c) Strain and angle are also dimensionless

Solid angle is dimensionless.

2. (d) $(N + 1)$ vernier scale division (VSD) = (N) main scale division

$$1 \text{ VSD} = \left(\frac{N}{N+1} \right) \text{MSD}$$

L.C = Least count = 1 MSD – 1 VSD

$$= \left(1 - \frac{N}{N+1} \right) \text{MSD}$$

$$= \left(\frac{1}{N+1} \right) \text{MSD}$$

$$= \left(\frac{1}{N+1} \right) \times \frac{0.1}{10} \text{ cm}$$

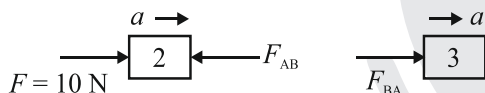
$$= \frac{1}{100(N+1)} \text{ cm}$$

3. (a) In connected motion,

$$\text{Common acceleration } a = \frac{F_{\text{net}}}{M_{\text{total}}}$$

$$a = \frac{10}{5} = 2 \text{ m/s}^2$$

Now, from FBD of 2 kg block & 3 kg block



Force exerted by block A on block B,

$$F_{BA} = (3)a$$

$$F_{BA} = 3 \times 2 = 6 \text{ N}$$

4. (b) A particle moving with uniform speed in a circular path has varying velocity as the direction of motion changes continuously. The magnitude of velocity remains constant. As the direction of centripetal acceleration is changing, the particle has varying acceleration.

5. (d) $T = M\omega^2 r$

$$T \propto \omega^2$$

T becomes 4 times

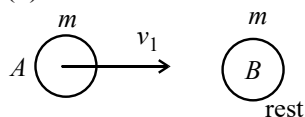
6. (c) $s = 2t - 1$

$$v = \frac{ds}{dt} = \frac{d}{dt}(2t - 1) = 2 \text{ (SI unit)}$$

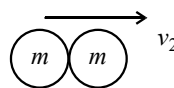
$$\text{Instantaneous power, } P = \vec{F} \cdot \vec{v}$$

$$= 5 \times 2 = 10 \text{ (SI unit)}$$

7. (d) Before collision:



After collision:



Now, from conservations of momentum,

Momentum before collision = Momentum after collision.

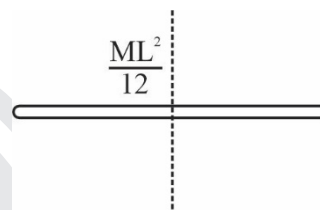
$$mv_1 + (0) = 2m v_2$$

$$mv_1 = 2mv_2$$

$$\frac{v_1}{v_2} = \frac{2}{1}$$

$$\therefore v_1 : v_2 = 2 : 1$$

8. (c)



$$\frac{ML^2}{12} = 2400$$

$$\Rightarrow \frac{400L^2}{12} = 2400$$

$$\Rightarrow L^2 = \frac{2400 \times 12}{400}$$

$$\Rightarrow L^2 = 72$$

$$\Rightarrow L = \sqrt{72}$$

$$\Rightarrow L = 8.5 \text{ cm}$$

9. (d) In case of rolling, the velocity of the point of contact with the surface is zero.

Also, the velocity of the point at the highest point of the rolling body is twice the velocity of COM of the body.

So, point P moves faster than point Q.

10. (b) $g = \frac{GM}{R^2} = 9.8 \text{ m/s}^2$... (i)

Now for Planet,

$$M' = \frac{M}{10}, D' = \frac{D}{2} \text{ [D is diameter of earth]}$$

$$\therefore R' = \frac{R}{2}$$

$$\text{Now, } g' = \frac{GM'}{(R')^2} = \frac{GM \times 4}{10 \times R^2} = \frac{4}{10} \times \frac{GM}{R^2}$$

Substitute value of $\frac{GM}{R^2}$ from equation (i)

$$g' = 0.4 \times 9.8 \text{ m/s}^2 = 3.92 \text{ m/s}^2$$

11. (c) $(\text{Stress})_{\max} = \text{Young's modulus} \times (\text{Strain})_{\max}$

$$\frac{8 \times 10^8}{2 \times 10^{11}} = (\Delta \ell)_{\max}$$

$$4 \times 10^{-3} = (\Delta \ell)_{\max}$$

$$(\Delta \ell)_{\max} = 4 \text{ mm}$$

12. (c) For circular disc,

$$F_{\text{excess}} = 2\pi r T$$

where r is radius of disc and T is surface tension.

Substituting the given values,

$$F_{\text{excess}} = 2\pi \times 4.5 \times 10^{-2} \times 0.07$$

$$= 2 \times \frac{22}{7} \times 4.5 \times 10^{-2} \times 7 \times 10^{-2}$$

$$= 198 \times 10^{-4} \text{ N} = 19.8 \text{ mN}$$

13. (c) As volume is constant along path bc , hence the work done, $W = \int P dV = 0$

14. (d) $x = 5 \sin \left(\pi t + \frac{\pi}{3} \right) \text{ m}$

$$A = 5 \text{ m}$$

$$\omega = \pi = \frac{2\pi}{T}$$

$$T = 2 \text{ s}$$

15. (a) Formula of potential is given by $V = \frac{Kp \cos \theta}{r^2}$

For axial point $\theta = 0^\circ$ or 180°

$$V = \pm \frac{Kp}{r^2} = \pm \frac{1}{4\pi\epsilon_0} \times \frac{p}{r^2}$$

$$= \pm \frac{9 \times 10^9 \times 4 \times 10^{-6}}{4}$$

$$= \pm 9 \times 10^3$$

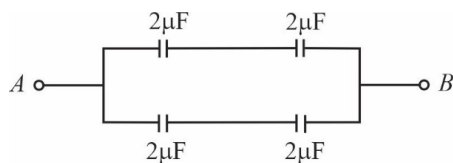
16. (b) For spherical shell

$$V_{\text{centre}} = V_{\text{surface}}$$

Difference in potential is zero.

17. (c) Balanced Wheatstone bridge

Equivalent circuit



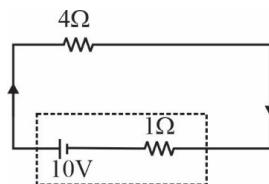
18. (d) Resistance of each part = 10Ω

First five parts equivalent = 50Ω

Next five parts equivalent = 2Ω

Again for series combination $R_{\text{eq}} = 52\Omega$

19. (a)



$$10 = 5 i$$

$$i = 2 \text{ A}$$

$$\text{Terminal voltage} = 10 - 1(2)$$

$$= 8 \text{ V}$$

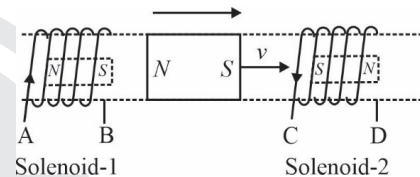
20. (a) Magnetic field $B = \frac{\mu_0 NI}{2r}$
- $$= \frac{4\pi \times 10^{-7} \times 100 \times 7}{2 \times 100 \times 10^{-2}} = 0.0044 \text{ T}$$

$$= 4.4 \text{ mT}$$

21. (c)

Material	Magnetic susceptibility (χ)
Diamagnetic	-10^{-5} to 10^{-9}
Paramagnetic	10^{-5} to 10^{-3}
Ferromagnetic	$\gg 1$
Non-magnetic	0

22. (c) Lenz Law



23. (b) $B = 0.049 \text{ T} = 4.9 \times 10^{-2} \text{ T}$

$$T = \frac{5}{20} = \frac{1}{4} \text{ s}, I = 9.8 \times 10^{-6} \text{ kg m}^2$$

$$T = 2\pi \sqrt{\frac{I}{MB}} \Rightarrow T^2 = \frac{4\pi^2 I}{MB}$$

$$M = \frac{4\pi^2 I}{T^2 B}$$

$$= \frac{4\pi^2 \times 9.8 \times 10^{-6}}{\left(\frac{1}{4}\right)^2 \times 4.9 \times 10^{-2}} = 1280\pi^2 \times 10^{-5} \text{ Am}^2$$

$$\therefore x = 1280 \pi^2$$

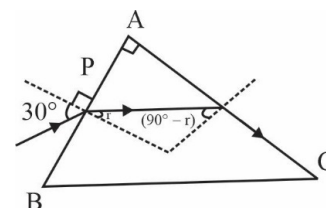
24. (d) $\frac{N_P}{N_S} = \frac{1}{2}$ (given)

We know that transformation ratio (r) is $r = \frac{N_S}{N_P} = \frac{V_S}{V_P}$

$$\therefore \frac{V_S}{V_P} = \frac{2}{1}$$

$$V_S : V_P = 2 : 1$$

25. (d)



For surface AB

$$1 \sin 30^\circ = \mu \sin r \quad \dots(1)$$

For surface AC

$$\mu \sin (90^\circ - r) = 1 \quad \dots(2)$$

$$\Rightarrow \mu \cos r = 1$$

$$\Rightarrow \cos r = \frac{1}{\mu}$$

From (1) and (2)

$$\Rightarrow \frac{1}{2} = \mu \sqrt{1 - \frac{1}{\mu^2}}$$

$$\Rightarrow \frac{1}{4} = \mu^2 \left(1 - \frac{1}{\mu^2}\right)$$

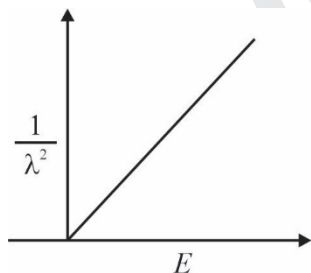
$$\Rightarrow \frac{1}{4} = \mu^2 - 1 \Rightarrow \mu^2 = 1 + \frac{1}{4}$$

$$\Rightarrow \mu = \frac{\sqrt{5}}{2}$$

26. (a) When we use white light in place of monochromatic light, the central fringe appears white, while the other fringes display different colours. This is because the central fringe for all colours is formed at central point, so it appears white after mixing up. As fringes of other colours fall at different positions, we see fringes of all colours on either side of the central bright fringe.

27. (b) At Brewster angle, only the reflected light is completely polarised, but the refracted light is partially polarised.

28. (b)



$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda^2 = \frac{h^2}{2mE}$$

$$E \propto \frac{1}{\lambda^2}$$

29. (d) Energy $E = h\nu$

$$\text{Velocity} = C = \nu\lambda$$

$$\lambda = \frac{c}{\nu}$$

$$P = \frac{h}{\lambda} = \frac{h}{\frac{c}{\nu}} = \frac{h\nu}{c}$$

$$F_{\text{ext}} = 0, \text{ So momentum conserved energy conserved.}$$

30. (a) Statement I is correct and II is incorrect.

Atoms are electrically neutral as they have equal number of positive and negative charges. Also, atoms of most elements are stable and they emit characteristics spectrum of their own.

31. (d)

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\lambda = \frac{1}{RZ^2} \left(\frac{n_1^2 n_2^2}{n_2^2 - n_1^2} \right)$$

For $H, Z = 1$ So,

$$\lambda = 91.2 \left(\frac{n_1^2 n_2^2}{n_2^2 - n_1^2} \right) \text{ nm}$$

$$\text{For } A \quad \lambda = 656.3$$

$$\text{For } B \quad \lambda = 486.1$$

$$\text{For } C \quad \lambda = 434.1$$

$$\text{For } D \quad \lambda = 410.2$$

32. (b) ${}^{290}_{82}X \xrightarrow{\alpha} {}^{286}_{80}Y \xrightarrow{e^+} {}^{286}_{79}Z \xrightarrow{\beta^-} {}^{286}_{80}P \xrightarrow{e^-} {}^{286}_{81}Q$

For α emission, mass number decreases by 4 and atomic number decreases by 2.

For e^+ , atomic number decreases by 1 and mass number remains unchanged.

For β^- , mass number is unchanged and atomic number increases by 1.

33. (c) The $I - V$ characteristics of solar cell is drawn in the fourth quadrant of the coordinate axes. This is because a solar cell does not draw current but supplies the same to the load.

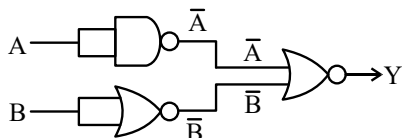
In reverse bias condition, the positive terminal is connected to the n -type and the negative terminal is connected to the p -type. Hence, majority charge carriers are swept into terminals and minority charge carriers are repelled by the terminals. Thus, a very small drift current flows through the diode due to minority charge carriers of both n -side and p -side.

34. (a) Output for given options.

(1) $\bar{B}$	(2) B	(3) $A.B + \bar{A}$	(4) $A.\bar{B} + \bar{A}$
1	0	1	1
0	1	1	1
1	0	0	1
0	1	1	0

So option (1) is correct.

35. (b)



$$Y = \overline{A + B} = \overline{A} \cdot \overline{B} \text{ (AND gate)}$$

36. (d) $F = \alpha t^2 + \beta t$

$$[MLT^{-2}] = \alpha [T^2]$$

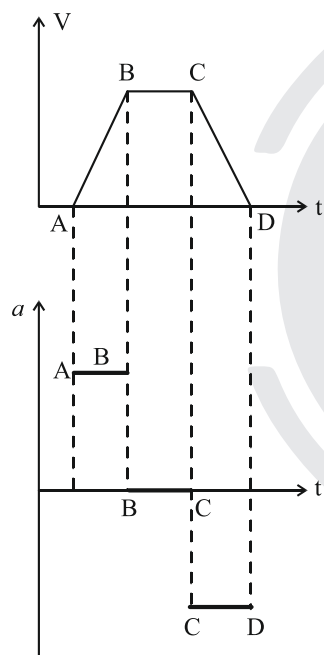
$$\alpha = [MLT^{-4}]$$

$$[MLT^{-2}] = \beta [T]$$

$$\beta = [MLT^{-3}]$$

$$\frac{\alpha t}{\beta} = \frac{[MLT^{-4}][T]}{[MLT^{-3}]} = [M^0L^0T^0]$$

37. (a)



a = slope of $v-t$ graph

from A to B → slope is positive and constant

→ so acceleration is positive & constant

B to C → slope is zero ⇒ acceleration is zero.

C to D → slope is negative and constant → so acceleration is negative & constant

38. (c) $(T.E)_{\text{initial}} = \text{PE of satellite}$

$$= \frac{-GMm}{R}$$

$(T.E)_{\text{final}} = \text{P.E of satellite} + \text{K.E of satellite}$

$$= \frac{-GMm}{3R} + \frac{1}{2}m\left(\sqrt{\frac{GM}{3R}}\right)^2$$

$$= \frac{-GMm}{6R}$$

Minimum energy required

$$= \frac{-GMm}{6R} + \frac{GMm}{R}$$

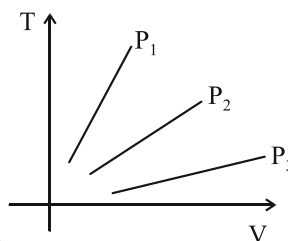
$$= \frac{-GMm + 6GMm}{6R} = \frac{5GMm}{6R}$$

39. (d) Thermal stress = thermal strain $\times$ Y

$$\frac{F}{A} = (\alpha \Delta T) Y$$

$$\frac{F}{10^{-3}} = (10^{-5}) \times 100 \times 0.5 \times 10^{11}$$

$$F = 50 \times 10^3 \text{ N}$$

40. (b) According to Charle's law $V \propto T$ 

$$P_1 > P_2 > P_3$$

41. (d) $m' = 3m$

$$l' = \frac{1}{2}l$$

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T' = 2\pi\sqrt{\frac{l'}{g}} = 2\pi\sqrt{\frac{l}{2g}}$$

$$T' = \left(\frac{1}{\sqrt{2}}\right) 2\pi\sqrt{\frac{l}{g}} = \frac{1}{\sqrt{2}}T$$

$$T' = \frac{x}{2}T$$

$$\frac{1}{\sqrt{2}}T = \frac{x}{2}T$$

$$x = \frac{2}{\sqrt{2}} = \sqrt{2}$$

42. (d) Keeping the battery connected

V remains same

$$C = \frac{\epsilon_0 A}{d}, d \text{ is reduced}$$

∴ C increases

From

$$Q = CV$$

As $C \uparrow$ $Q \uparrow \Rightarrow A$ is correct

$$E = \frac{1}{2}CV^2 \text{ As } C \uparrow E \text{ increases} \Rightarrow B \text{ is incorrect}$$

Capacitance increases $\Rightarrow C$ is correct

Product of charge and voltage increases

$\Rightarrow E$ is correct.

43. (c) In option 3 if the diode is considered non ideal with resistance 10Ω then the bridge can be balanced.

44. (d) $P_1 = 1\text{kW}$, $P_2 = 2\text{kW}$

For parallel combination

$$P_{\text{parallel}} = P_1 + P_2 = 1 + 2 = 3\text{kW}$$

For series combination

$$\frac{1}{P_{\text{series}}} = \frac{1}{P_1} + \frac{1}{P_2}$$

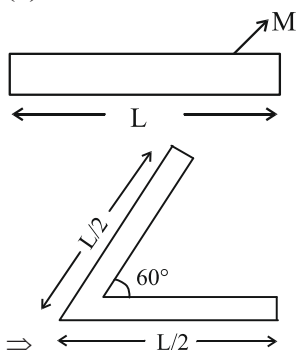
$$= 1 + \frac{1}{2}$$

$$P_{\text{series}} = \frac{2}{3}\text{kW}$$

$$\frac{P_{\text{parallel}}}{P_{\text{series}}} = \frac{3}{\frac{2}{3}} \times 3 = \frac{9}{2}$$

$$\frac{P_{\text{series}}}{P_{\text{parallel}}} = \frac{2}{9}$$

45. (d) $M = mL$



Magnetic pole strength remains the same

$$M_{\text{new}} = mL_{\text{eff}} = \frac{mL}{2} = \frac{M}{2} \text{ (effective is distance between ends)}$$

46. (d) If sheet is magnetic, then force is required to hold the sheet because magnetic pole exert force on sheet

As sheet is conducting, eddy current formed which oppose the way by which it is produced, so force is required.

As sheet is non conducting & non polar, so no magnetization occur in sheet hence no force is required.

47. (d) $V_{\text{rms}} = 210\text{ volt}$,

$$V_0 = 210\sqrt{2}\text{ volt}$$

$$V_0 = I_0 X_C$$

$$V_0 = I_0 \times \frac{1}{\omega C}$$

$$210\sqrt{2} = I_0 \times \frac{1}{2\pi \times 50 \times 10 \times 10^{-6}}$$

$$I_0 = 210\sqrt{2} \times 2\pi \times 500 \times 10^{-6}$$

$$I_0 = 0.932\text{A}$$

48. (b) EM waves do not originate from charges moving with uniform speed.

49. (d) As capacitor is being charged the displacement current is

$$I = \epsilon_0 \frac{d(\phi)_E}{dt}$$

$$I = \epsilon_0 \frac{d}{dt}(EA)$$

$$I = \epsilon_0 \frac{d}{dt}\left(\frac{V}{d}A\right)$$

$$I = \frac{\epsilon_0 A}{d} \frac{d}{dt}\left(\frac{Q}{C}\right)$$

$$I = \frac{dQ}{dt} = I_{\text{conduction}}$$

50. (d) $f_0 = 140\text{ cm}$, $f_e = 5\text{ cm}$

For normal adjustment

$$M = \left| \frac{-f_0}{f_e} \right| = \left| \frac{-140}{5} \right| = 28$$